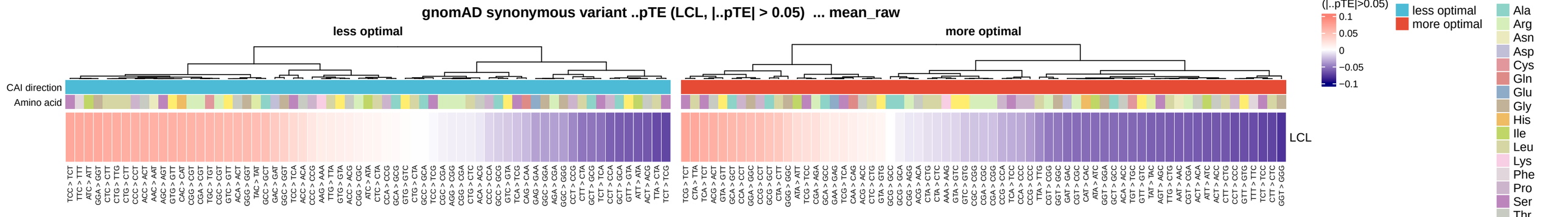
gnomAD synonymous variant ..pTE (LCL, |..pTE| > 0.05) ... mean_raw
less optimal
more optimal
CAI direction
Amino acid
LCL
(|..pTE|>0.05)
0.1
0.05
0
−0.05
−0.1
less optimal
more optimal
Ala
Arg
Asn
Asp
Cys
Gln
Glu
Gly
His
Ile
Leu
Lys
Phe
Pro
Ser
Thr
TCC > TCT
TTC > TTT
ATC > ATT
GGA > GGT
CTC > CTT
CTG > CTT
CCC > CCT
ACC > ACT
AAC > AAT
AGC > AGT
GTG > GTT
CAC > CAT
CGG > CGT
CGA > CGT
TGC > TGT
CGC > CGT
GTC > GTT
ACA > ACT
GGG > GGT
TAC > TAT
GCC > GCT
GAC > GAT
GGC > GGT
TCC > TCA
ACC > ACA
CCC > CCG
AAG > AAA
TTG > TTA
GTG > GTA
ACC > ACG
CGG > CGC
ATC > ATA
CTC > CTA
CCA > CCG
GCA > GCG
GTG > GTC
CTG > CTA
GCC > GCA
TCC > TCG
CGC > CGA
AGG > CGG
CGG > CGA
CTG > CTC
ACA > ACG
CCC > CCA
GCC > GCG
GTC > GTA
TCA > TCG
CAG > CAA
GAG > GAA
GGC > GGA
AGA > CGA
GGC > GGG
CCT > CCG
CTT > CTA
GCT > GCG
TCT > TCA
CCT > CCA
GCT > GCA
GTT > GTA
ATT > ATA
ACT > ACG
TTA > CTA
TCT > TCG
TCG > TCT
CTA > TTA
TCA > TCT
ACG > ACT
GTA > GTT
GCA > GCT
CCA > CCT
GGA > GGC
CCG > CCT
GCG > GCT
CTA > CTT
GGG > GGC
ATA > ATT
TCG > TCC
CGA > AGA
GCA > GCC
GAA > GAG
TCG > TCA
CAA > CAG
ACG > ACC
CTC > CTG
GTA > GTG
GCG > GCC
GCG > GCA
CGG > AGG
ACG > ACA
CTA > CTC
CTA > CTG
AAA > AAG
GTA > GTC
GTC > GTG
CGC > CGG
CGA > CGC
CCG > CCA
TCA > TCC
CCA > CCC
CCG > CCC
TTA > TTG
CGT > CGG
GGT > GGC
GAT > GAC
CGT > CGC
CAT > CAC
ATA > ATC
GGT > GGA
GCT > GCC
ACA > ACC
TGT > TGC
GTT > GTC
TAT > TAC
AGT > AGC
TTG > CTG
AAT > AAC
CGT > CGA
ACT > ACA
ATT > ATC
ACT > ACC
CTT > CTG
CCT > CCC
GTT > GTG
TTT > TTC
TCT > TCC
CTT > CTC
GGT > GGG